Hospital-Level Variation in Liver Radiodensity and Chemotherapy-Induced Toxicities in Colon Cancer Patients

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1 Introduction

Capecitabine and oxaliplatin are commonly used chemotherapeutic agents in the treatment of stage II and stage III colorectal cancer, either as monotherapy or in combination (CAPOX regimen). Although effective, both agents are associated with clinically relevant toxicities that may lead to dose reductions, treatment delays, or early discontinuation. Capecitabine-related toxicities frequently include hand-foot syndrome and gastrointestinal side-effects (Scheithauer et al., 2003; Iacovelli et al., 2014). In contrast, oxaliplatin is particularly associated with peripheral neuropathy, a dose-limiting toxicity that develops in a large proportion of treated patients and may persist over time (Renting et al., 2024). It is therefore critical to identify factors that influence treatment-related toxicities, and this remains an important clinical and research priority.

Recent analyses within the COLON cohort, a prospective multicentre cohort study conducted in the Netherlands, showed that circulating folic acid and folic acid supplement use were associated with an increased risk of capecitabine-induced toxicities among patients with colon or rectal cancer (Kok et al., 2024). Additional work within this cohort has examined nutritional determinants of chronic oxaliplatin-induced toxicities. These findings suggest that metabolic and nutritional factors may predispose patients to chemotherapy-induced toxicities, although causal relations cannot be established within observational designs.

In the current project, we focus on liver fat accumulation, measured as liver radiodensity on computed tomography (CT) scans taken at the time of diagnosis, as a potential determinant of chemotherapy-induced side-effects. The data originate from the prospective COLON study, in which patients newly diagnosed with colorectal cancer were recruited from 11 hospitals in the Netherlands (Winkels et al., 2014). While inclusion of patients from different hospitals improves generalisability, it also introduces potential heterogeneity. Differences in CT scanner characteristics, imaging protocols, and calibration procedures across hospitals may lead to systematic variation in measured liver radiodensity. In addition, differences in patient characteristics across hospitals (such as age, tumour stage, or general health status) or differences in clinical practices across hospitals may influence the occurrence of treatment-related toxicities.

If hospital-level differences are present, they may affect the estimated association between liver radiodensity and risk of chemotherapy-induced toxicities. Ignoring such variation could lead to biased estimates or misleadingly precise confidence intervals. Therefore, it is important to assess whether hospital-level effects exist and to determine how they influence statistical inference. The primary aim of this paper is to evaluate how hospital-level variation influences the estimated risk and timing of chemotherapy-induced toxicities in relation to liver radiodensity at diagnosis within a multicentre observational cohort. Specifically, we address the following main research question:

RQ: *How does hospital-level variation influence the estimated association between liver radiodensity and time to chemotherapy-induced toxicity?*

The study specifically answers the following sub-questions:

- **SQ1.** Are there systematic differences between hospitals in liver radiodensity measurements and/or toxicity outcomes?
- **SQ2.** How does the estimated association between liver radiodensity and time to toxicity change when hospital-level variation is accounted for, compared to ignoring it?
- **SQ3.** Given any observed hospital heterogeneity, what does the within-hospital association imply about liver radiodensity as a predictor of chemotherapy-induced toxicity at clinically realistic exposure scales?

2 Methods

This section describes the study design, variables, and statistical approach used to investigate the association between liver radiodensity and time to chemotherapy-induced toxicity. Particular attention is given to the multicentre nature of the data and the potential role of hospital-level variation.

2.1 Participants and Study Design

The data originate from a prospective multicentre cohort study conducted in the Netherlands, in which patients diagnosed with colorectal cancer were recruited from multiple hospitals and followed during chemotherapy treatment. The study has an observational design, and no intervention was assigned as part of this analysis. Patients were included if they received chemotherapy involving capecitabine, oxaliplatin, or a combination of both (CAPOX regimen), and if computed tomography (CT) scans at the time of diagnosis were available. Because not all patients received the same treatment, separate analytic samples were defined for each toxicity endpoint. Patients were excluded from a given analysis if they were not at risk for the corresponding toxicity or if information on time to toxicity or event occurrence was missing.

2.2 Measurements

Outcome: Time to Dose-Limiting Toxicity. The primary outcome is time to first occurrence of dose-limiting toxicity during chemotherapy treatment. Time was operationalised as the number of chemotherapy cycles completed before the onset of toxicity. For each endpoint, the outcome consists of a time variable and an event indicator specifying whether toxicity occurred. Patients who did not experience toxicity during the observed treatment period were treated as censored at their last completed cycle. Three toxicity outcomes were considered: toxicity associated with capecitabine (CAP), toxicity associated with oxaliplatin (OXA), and toxicity associated with the combined CAPOX regimen. Dose-limiting toxicity was defined as any toxicity event leading to a dose reduction, treatment delay, or premature discontinuation of chemotherapy. Each outcome defines a separate risk set, as only patients receiving the corresponding treatment are at risk for that specific toxicity.

Exposure: Liver Radiodensity. The main exposure is liver radiodensity measured on CT scans at the time of diagnosis. Radiodensity is expressed in Hounsfield Units (HU), with lower values indicating higher liver fat content. Liver radiodensity was analysed as a continuous variable.

Covariates. The analyses adjust for a set of patient and treatment characteristics that may influence the risk of chemotherapy-induced toxicity, including age at diagnosis, sex, body mass index, smoking status, and starting dose of chemotherapy. Starting dose was defined in a treatment-specific manner depending on the toxicity endpoint considered: the capecitabine starting dose was used for the CAP endpoint, the oxaliplatin starting dose for the OXA endpoint, and both doses were included simultaneously for the CAPOX endpoint.

Hospital. Patients were recruited from multiple hospitals. Hospital was treated as a grouping variable to account for potential differences in imaging procedures, patient characteristics, and clinical practices across centres.

2.3 Statistical Analysis

Descriptive Analysis. Descriptive statistics were used to summarise the study population and to explore the distribution of toxicity outcomes across endpoints and hospitals. Event rates were calculated for each toxicity endpoint, and variation in toxicity occurrence across hospitals was examined.

Differences in liver radiodensity across hospitals were assessed using one-way analysis of variance (ANOVA). This analysis evaluates whether mean radiodensity values differ between hospitals and provides insight into potential measurement heterogeneity across centres.

Test for hospital differences in toxicity rates. To determine whether the observed differences in event rates across hospitals exceeded what binomial sampling alone would produce, a Pearson's chi-square test of independence was applied to the hospital $\times$ event contingency table for each endpoint. A non-significant result indicates that the observed range of hospital event rates is consistent with sampling variation around a common rate.

Survival Analysis. The association between liver radiodensity and time to dose-limiting toxicity was analysed using Cox proportional hazards regression models. Separate models were fitted for each toxicity endpoint (CAP, OXA, and CAPOX), using endpoint-specific datasets that include only patients at risk for the respective toxicity. The primary exposure in all models was liver radiodensity. Models were adjusted for age, sex, body mass index, smoking status, and treatment-specific starting dose.

Cox proportional hazards models estimate the association between one or more variables and the time until an event occurs, in this case dose-limiting toxicity. The model is based on the concept of the hazard: the rate at which the event occurs at any given point in time, among patients who have not yet experienced it. The main output is the hazard ratio (HR). A value of 1.0 indicates no association; values above 1.0 indicate a higher rate of toxicity and values below 1.0 a lower rate. A key advantage of Cox models is the correct handling of censoring: patients who complete treatment without experiencing toxicity, or who leave follow-up early, are included in the analysis rather than excluded, and their observation time contributes up to the point they are last observed.

To address the role of hospital-level variation, two model specifications were fitted for each endpoint: an unstratified Cox model (no adjustment for hospital), and a stratified Cox model in which each hospital was assigned its own baseline hazard function. In the stratified specification the liver-radiodensity effect is estimated entirely from within-hospital comparisons, so any systematic between-hospital differences in CT calibration, case-mix, or clinical practice are absorbed by the hospital-specific baselines and cannot bias the estimate. Side-by-side comparison of the two specifications quantifies the contribution of hospital-level confounding (SQ2).

Reporting effect sizes. Liver radiodensity in this cohort varies on a scale of tens of HU rather than single units, so a per-1-HU hazard ratio is too granular for meaningful clinical interpretation. The hazard ratio for liver radiodensity is therefore reported per 10 HU, which is conventional in CT-based clinical literature and makes the effect comparable across studies. A per-standard-deviation hazard ratio is also reported in the same table for context, since it reflects the typical exposure contrast actually observed in the cohort.

Model Assumptions. The proportional hazards assumption was assessed using Schoenfeld residual tests for all three stratified models. This assumption implies that the effect of liver radiodensity on toxicity risk remains constant over time. Missingness in covariate data was assessed prior to analysis. No imputation was performed, as the proportion of missing values was low and unlikely to meaningfully affect the results.

Software. All analyses were conducted using R (version 4.6.0; R Core Team), with survival models fitted via the `survival` package, model output processed using `broom`, and tables and figures produced with `kableExtra`, `ggplot2`, `survminer`, and `forestmodel`.

3 Results

3.1 Sample Characteristics and Event Rates

The analysis included patients receiving capecitabine (CAP), oxaliplatin (OXA), or a combination of both (CAPOX). Because not all patients received each treatment, separate datasets were constructed for each toxicity endpoint. The number of patients eligible for each endpoint (non-missing event indicator) was 344, 311, and 359 for CAP, OXA, and CAPOX respectively. After restricting to patients with complete covariate information required for the Cox models, the analytic samples comprised 314, 289, and 305 patients respectively. Table 1 presents the event counts for each analytic sample.

Table 1: Observed toxicity rates by endpoint (complete-case analytic sample)

Endpoint	N	Events	Censored	Event rate (%)
CAP	314	164	150	52.2
OXA	289	236	53	81.7
CAPOX	305	234	71	76.7

Across the three endpoints, dose-limiting toxicity was a common outcome. The proportion was highest for oxaliplatin-related toxicity. The number of observed events was sufficient to support time-to-event analyses for all endpoints.

3.2 Variation in Toxicity Across Hospitals

Observed toxicity rates varied across hospitals, with proportions ranging from roughly one-third to four-fifths across centres for the three endpoints. However, with relatively small numbers of patients per hospital, some of this variation can be expected from binomial sampling alone even when the underlying toxicity rates are identical. To distinguish real between-hospital differences from sampling noise, Pearson's chi-square test of independence was applied to the hospital $\times$ event contingency table for each endpoint (Table 2).

Table 2: Pearson's chi-square test of equal toxicity rates across hospitals, by endpoint. A small p-value indicates that observed between-hospital differences exceed what would be expected from binomial sampling alone.

Endpoint	N	Hospitals	Chi-sq	df	p-value
CAP	314	11	16.90	10	0.077
OXA	289	11	30.86	10	<0.001
CAPOX	305	11	15.16	10	0.126

The chi-square results reveal a more nuanced picture than the raw event-rate range alone suggests. For oxaliplatin-related toxicity (OXA), hospital event rates differ significantly ($p < 0.001$), indicating real between-hospital differences in OXA toxicity rates. For capecitabine (CAP, $p = 0.077$) and the combined regimen (CAPOX, $p = 0.126$), however, the observed range is consistent with sampling variation around a common rate and we cannot conclude that underlying toxicity rates differ between hospitals.

Even where the chi-square test does not detect between-hospital differences in event rates, hospital remains a methodologically important grouping variable. Differences in CT scanner calibration, imaging protocols, and patient case-mix can still bias estimates of the liver-radiodensity effect (see Section 3.3 for direct evidence on this point), independent of whether crude event rates differ.

3.3 Variation in Liver Radiodensity Across Hospitals

Visual inspection of liver radiodensity distributions indicated differences across hospitals (Figure 1). This was confirmed by a one-way ANOVA, which showed that mean liver radiodensity differed significantly between hospitals. Table 3 presents the ANOVA results.

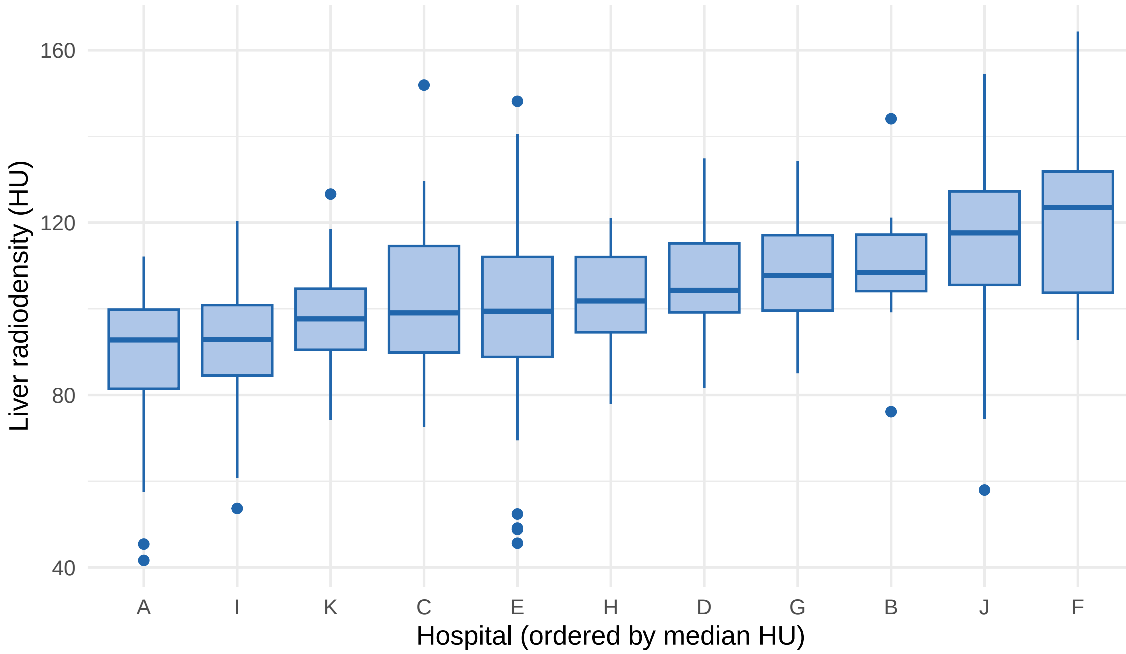

Figure 1: Distribution of liver radiodensity (Hounsfield Units) by hospital. Each box shows the interquartile range (IQR) and median for one hospital; whiskers extend to the most extreme value within 1.5 IQR of the box edges; points beyond are shown individually. Hospitals are ordered by median HU. Lower HU indicates higher liver fat content.

Table 3: One-way ANOVA testing whether mean liver radiodensity (HU) differs between hospitals. Df = degrees of freedom; Sum Sq = sum of squares; Mean Sq = mean square; F = F-statistic for between-hospital variability relative to within-hospital variability.

Source	Df	Sum Sq	Mean Sq	F	p-value
Hospital	10	32418.1	3241.8	12.07	<0.001
Residuals	348	93482.9	268.6		

These findings indicate that liver radiodensity measurements are not directly comparable across hospitals and may be influenced by differences in imaging procedures or patient populations. This suggests that hospital may act as a confounding factor in the association between liver radiodensity and toxicity.

3.4 Time to Toxicity

Kaplan-Meier curves were used to describe the unadjusted time to dose-limiting toxicity across chemotherapy cycles (Figure 2). Across all endpoints, the probability of remaining free from toxicity declined rapidly during the early treatment cycles, indicating that most toxicities occurred early in treatment. In later cycles, estimates became less stable due to the smaller number of patients remaining at risk. These patterns support the use of survival analysis methods and highlight the clinical relevance of early treatment tolerance.

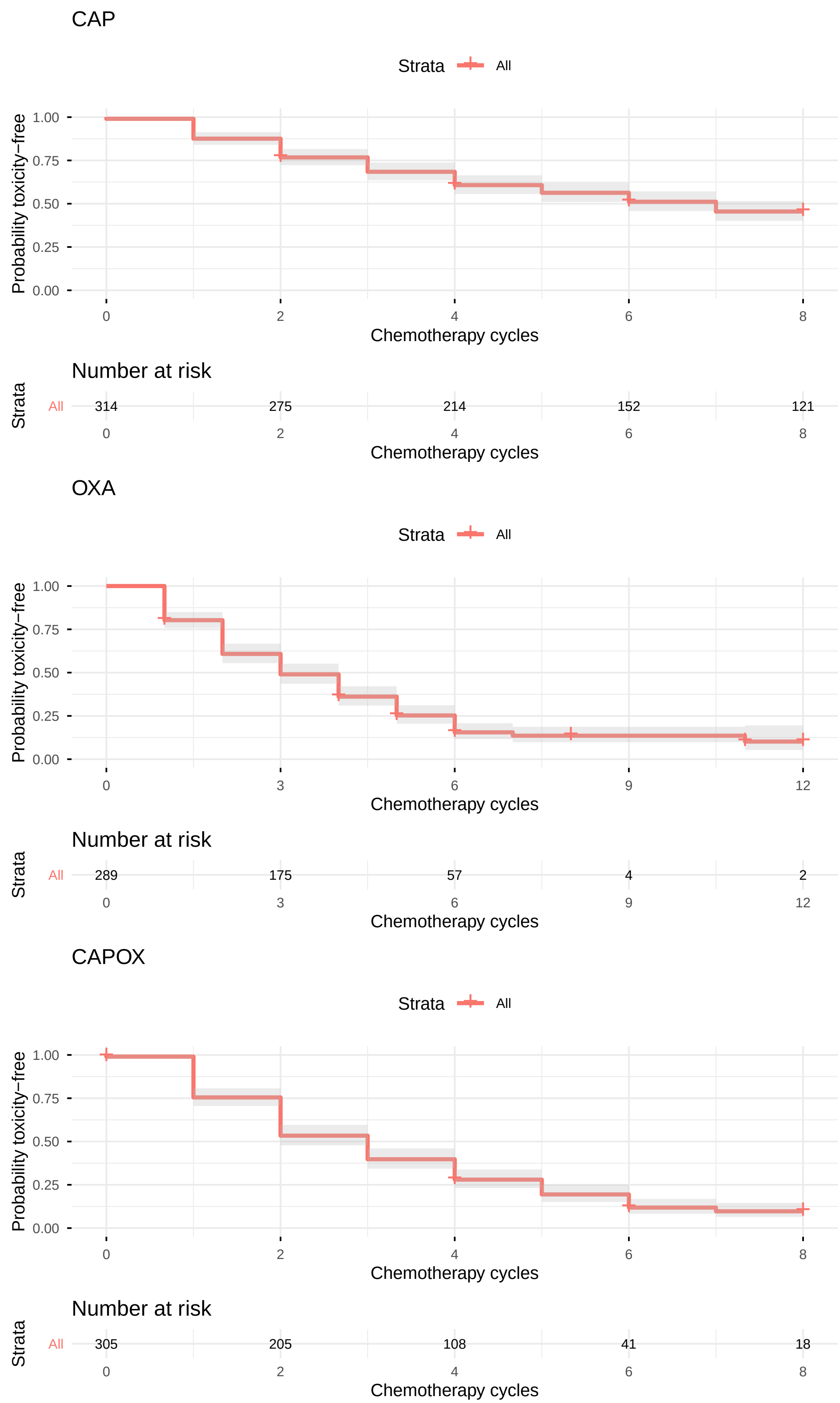

Figure 2: Kaplan-Meier curves for time to dose-limiting toxicity by endpoint. Shaded bands represent 95% confidence intervals. Risk tables show the number of patients remaining under observation at each cycle.

3.5 Association between Liver Radiodensity and Toxicity

To address the central question of how hospital-level variation influences the estimated liver-radiodensity effect (SQ2), Cox models were fitted in two specifications for each endpoint: an unstratified model (no adjustment for hospital), and a stratified model (each hospital with its own baseline hazard, so the liver-radiodensity coefficient is estimated entirely from within-hospital comparisons). Both specifications adjust for age, sex, body mass index, smoking status, and treatment-specific starting dose. Table 4 presents the liver-radiodensity hazard ratio under both specifications.

Table 4: Liver-radiodensity hazard ratio per 10 HU from unstratified and stratified Cox models, by endpoint. The stratified specification estimates the within-hospital effect; differences between the two specifications reflect the contribution of hospital-level confounding. All models additionally adjust for age, sex, BMI, smoking, and starting dose.

Endpoint	Model	HR	95% CI	p-value
CAP	Unstratified	1.06	0.97–1.16	0.195
CAP	Stratified by hospital	1.05	0.95–1.16	0.376
OXA	Unstratified	0.99	0.91–1.06	0.702
OXA	Stratified by hospital	1.09	1.00–1.20	0.060
CAPOX	Unstratified	1.00	0.93–1.08	0.994
CAPOX	Stratified by hospital	1.10	1.00–1.21	0.044

For the oxaliplatin-containing endpoints (OXA and CAPOX), the two specifications produce different results. The unstratified models show no association between liver radiodensity and toxicity, with hazard ratios close to 1.00 per 10 HU and p-values well above conventional thresholds. After stratifying by hospital, a within-hospital effect emerges: HR = 1.09 (95% CI 1.00–1.20, $p = 0.060$) for OXA and HR = 1.10 (95% CI 1.00–1.21, $p = 0.044$) for CAPOX. For CAP, the two specifications agree closely and neither indicates an association. The pattern suggests that, for OXA and CAPOX, ignoring hospital variation obscures the within-hospital association between liver radiodensity and toxicity. This is consistent with the hospital-level heterogeneity in liver radiodensity documented in Section 3.3, which likely reflects differences in CT-scanner calibration across centres.

3.5.1 Effect size at a per-SD scale

The per-10-HU hazard ratios reported above represent a fixed, clinically transparent exposure contrast that is comparable across studies. As a complementary summary, Table 5 reports the same stratified hazard ratios on a per-standard-deviation scale, where one unit corresponds to the typical between-patient contrast in liver radiodensity observed in this cohort.

For CAPOX, a one-SD difference in liver radiodensity (approximately 19.3 HU) is associated with a hazard ratio of 1.21 (95% CI 1.01–1.45). For OXA the per-SD HR is 1.19, borderline. The confidence intervals remain wide and the lower bound for CAPOX is close to 1, but the effect size at this scale is more readily interpretable than the per-1-HU framing might suggest.

Table 5: Liver-radiodensity hazard ratio from stratified Cox models at two exposure scales: per 10 HU and per one within-cohort standard deviation of liver HU. The HU-per-unit column shows the HU difference each scale represents.

Endpoint	Scale	HU per unit	HR	95% CI
CAP	per 10 HU	10.0	1.05	0.95–1.16
CAP	per 1 SD	19.2	1.09	0.90–1.33
OXA	per 10 HU	10.0	1.09	1.00–1.20
OXA	per 1 SD	18.9	1.19	0.99–1.41
CAPOX	per 10 HU	10.0	1.10	1.00–1.21
CAPOX	per 1 SD	19.3	1.21	1.01–1.45

3.5.2 Full adjusted model

Table 6 presents the complete set of adjusted hazard ratios from the stratified models, including all covariates.

Table 6: Full adjusted hazard ratios from stratified Cox models, by endpoint. All HRs reported with 2-decimal precision so that values close to 1 remain distinguishable. Reference categories: sex = female; smoking = category 1.

Covariate	HR (95\% CI)	p-value
CAP		
Liver HU (per 1 HU)	1.00 (0.99, 1.02)	0.376
Age (per year)	1.02 (1.00, 1.04)	0.074
Sex (male vs. female)	0.76 (0.54, 1.07)	0.120
BMI (per unit)	0.98 (0.94, 1.02)	0.371
Smoking (cat. 2 vs. 1)	1.33 (0.68, 2.59)	0.405
Smoking (cat. 3 vs. 1)	1.50 (0.77, 2.93)	0.238
Starting dose	1.00 (1.00, 1.00)	0.090
OXA		
Liver HU (per 1 HU)	1.01 (1.00, 1.02)	0.060
Age (per year)	1.03 (1.01, 1.05)	0.001
Sex (male vs. female)	0.48 (0.34, 0.68)	<0.001
BMI (per unit)	0.98 (0.94, 1.02)	0.278
Smoking (cat. 2 vs. 1)	1.98 (1.11, 3.56)	0.022
Smoking (cat. 3 vs. 1)	2.18 (1.21, 3.90)	0.009
Starting dose	1.01 (1.01, 1.02)	<0.001
CAPOX		
Liver HU (per 1 HU)	1.01 (1.00, 1.02)	0.044
Age (per year)	1.03 (1.01, 1.05)	0.010
Sex (male vs. female)	0.63 (0.45, 0.89)	0.008
BMI (per unit)	0.98 (0.94, 1.02)	0.331
Smoking (cat. 2 vs. 1)	1.83 (1.04, 3.22)	0.036
Smoking (cat. 3 vs. 1)	1.97 (1.11, 3.48)	0.020
Starting dose CAP	1.00 (1.00, 1.00)	0.968
Starting dose OXA	1.01 (1.00, 1.02)	0.056

For OXA and CAPOX, several covariates were significantly associated with toxicity (Table 6).

Older age, female sex (compared to male), and current smoking were each associated with a higher hazard of toxicity, with consistent patterns across the two oxaliplatin-containing regimens. Oxaliplatin starting dose was also associated with toxicity in the OXA model. None of these covariates reached statistical significance in the CAP model, suggesting that capecitabine-related toxicity is less strongly driven by the patient characteristics included in this analysis.

3.6 Model Diagnostics

The proportional hazards assumption was assessed using Schoenfeld residual tests for each stratified Cox model. Table 7 reports the test statistic, degrees of freedom, and p-values for all variables in each model. A non-significant p-value ($p > 0.05$) indicates no evidence against the proportional hazards assumption. The GLOBAL row tests the assumption across all covariates simultaneously.

Table 7: Schoenfeld residual tests: proportional hazards assumption (all models).

Variable	Chi-sq	df	p-value
CAP			
liverHU_manual	0.115	1	0.734
age	1.542	1	0.214
sex	1.386	1	0.239
BBMI	0.619	1	0.432
BSMOKER	0.635	2	0.728
starting_dose	1.677	1	0.195
GLOBAL	5.543	7	0.594
OXA			
liverHU_manual	2.530	1	0.112
age	0.024	1	0.876
sex	2.324	1	0.127
BBMI	1.187	1	0.276
BSMOKER	2.268	2	0.322
starting_dose	0.055	1	0.814
GLOBAL	9.333	7	0.230
CAPOX			
liverHU_manual	0.015	1	0.902
age	0.211	1	0.646
sex	1.336	1	0.248
BBMI	0.355	1	0.551
BSMOKER	0.634	2	0.728
doseCAP	1.031	1	0.310
doseOXA	0.108	1	0.742
GLOBAL	3.937	8	0.863

All p-values are well above the conventional threshold of 0.05, indicating no evidence that the proportional hazards assumption is violated for any variable or for any of the three models. These findings support the validity of the Cox model and suggest that the estimated hazard ratios can be interpreted as constant over the treatment period.

4 Discussion

This study examined how hospital-level variation in a multicentre observational cohort influences the estimated association between liver radiodensity and time to chemotherapy-induced toxicity. The three sub-questions from the introduction are addressed in turn below.

4.1 Hospital differences in measurements and outcomes (SQ1)

Liver radiodensity differs significantly between hospitals (Table 3, $p < 0.001$), consistent with systematic between-centre differences in CT-scanner calibration, imaging protocols, or patient case-mix.

Hospital differences in toxicity rates are more nuanced. Pearson's chi-square test (Table 2) indicates a significant difference for OXA ($p < 0.001$), but not for CAP ($p = 0.077$) or CAPOX ($p = 0.126$). For these two endpoints, the observed range of hospital event rates is consistent with binomial sampling variation around a common rate. The descriptive observation that proportions span a wide range across centres is therefore not, on its own, sufficient evidence of underlying heterogeneity.

4.2 How accounting for hospitals changes the estimate (SQ2)

The comparison of unstratified and stratified Cox models (Table 4) shows that hospital-level variation has a clear effect on the estimated association for the oxaliplatin-containing endpoints. In the unstratified models, the hazard ratios for OXA and CAPOX are close to 1.00 per 10 HU and not statistically significant. After stratification, a within-hospital effect is recovered: HR = 1.09 (95% CI 1.00–1.20, $p = 0.060$) for OXA and HR = 1.10 (95% CI 1.00–1.21, $p = 0.044$) for CAPOX. For CAP the two specifications agree closely, and neither suggests an association.

The interpretation is that ignoring hospital variation does not just shift the estimate, it obscures the within-hospital association entirely for OXA and CAPOX. The likely mechanism is the systematic between-hospital difference in measured liver radiodensity (Section 3.3): because measured HU is not directly comparable across centres, an unstratified model mixes within- and between-hospital contrasts in a way that washes out the within-hospital signal. Stratification by hospital removes this mixing.

4.3 What the within-hospital association implies (SQ3)

For CAPOX, a difference of 10 HU in liver radiodensity is associated with a hazard ratio of 1.10 (95% CI 1.00–1.21); on a per-SD scale (approximately 19.3 HU) the same effect corresponds to a hazard ratio of 1.21. For OXA the per-10-HU hazard ratio is 1.09, borderline. These are modest effect sizes, but they are no longer trivially close to 1 once the exposure scale reflects the variation actually observed in the cohort.

Several caveats apply. The lower bound of the CAPOX 95% confidence interval is close to 1, so the data are also compatible with a substantially weaker effect. The CAP endpoint shows no association at any scale and offers no evidence for liver radiodensity as a predictor of capecitabine-only toxicity. Finally, the within-hospital association is recovered only after stratification removes between-hospital measurement and case-mix differences; comparisons across centres would not be interpretable in the same way, because measured liver radiodensity is not directly comparable across hospitals.

4.4 Limitations

The analysis is observational, and a causal interpretation of the within-hospital association is not warranted. Liver radiodensity may proxy for unmeasured nutritional, metabolic, or anthropometric factors that themselves drive toxicity risk. Statistical power is modest, particularly for the CAP endpoint, and the wide confidence intervals at clinically meaningful exposure scales are a direct consequence. The proportional hazards assumption was not violated for any endpoint (Section 3.6), but the analysis cannot rule out non-linearity in the liver-radiodensity effect. Future analyses could consider categorical specifications or splines to assess whether risk is concentrated at clinically defined thresholds (for example, liver HU below 40, indicating moderate to severe steatosis).

4.5 Conclusion

After accounting for hospital-level variation through stratification, liver radiodensity shows a small but consistent within-hospital association with time to dose-limiting toxicity for the oxaliplatin-containing regimens (OXA and CAPOX), reaching conventional significance for CAPOX. The same association is not detected in unstratified models, indicating that hospital-level variation is an important feature of the data rather than an analytic detail. Liver radiodensity remains an exploratory predictor for this outcome, and additional evidence would be needed before it could be considered for clinical use.

5 References

Iacovelli, R., Pietrantonio, F., Palazzo, A., Maggi, C., Ricchini, F., de Braud, F., et al. (2014). Incidence and relative risk of grade 3 and 4 diarrhoea in patients treated with capecitabine or 5-fluorouracil: A meta-analysis of published trials. *British Journal of Clinical Pharmacology, 78*(6), 1228–1237.

Kok, D. E., van Duijnhoven, F. J. B., Lubberman, F. J. E., McKay, J. A., van Lanen, A., Winkels, R. M., et al. (2024). Intake and biomarkers of folate and folic acid as determinants of chemotherapy-induced toxicities in patients with colorectal cancer: A cohort study. *The American Journal of Clinical Nutrition, 119*, 294–301.

Renting, L., Zwart, N. R. K., Ueland, P. M., McCann, A., Ulvik, A., van Halteren, H. K., Lubberman, F. J. E., Winkels, R. M., Kampman, E., & Kok, D. E. (2024). Vitamin B6 status and chronic chemotherapy-induced peripheral neuropathy: a prospective cohort study among patients with non-metastatic colorectal cancer receiving oxaliplatin-based chemotherapy. *BMJ Oncology, 3*: e000462.

Scheithauer, W., McKendrick, J., Begbie, S., Borner, M., Burns, W. I., Burris, H. A., et al. (2003). Oral capecitabine as an alternative to intravenous 5-fluorouracil-based adjuvant therapy for colon cancer: Safety results of a randomized phase III trial. *Annals of Oncology, 14*(12), 1735–1743.

Winkels, R. M., Heine-Bröring, R. C., van Zutphen, M., van Harten-Gerritsen, S., Kok, D. E., van Duijnhoven, F. J. B., et al. (2014). The COLON study: Colorectal cancer: Longitudinal, Observational study on Nutritional and lifestyle factors that may influence colorectal tumour recurrence, survival and quality of life. *BMC Cancer, 14*, 374.